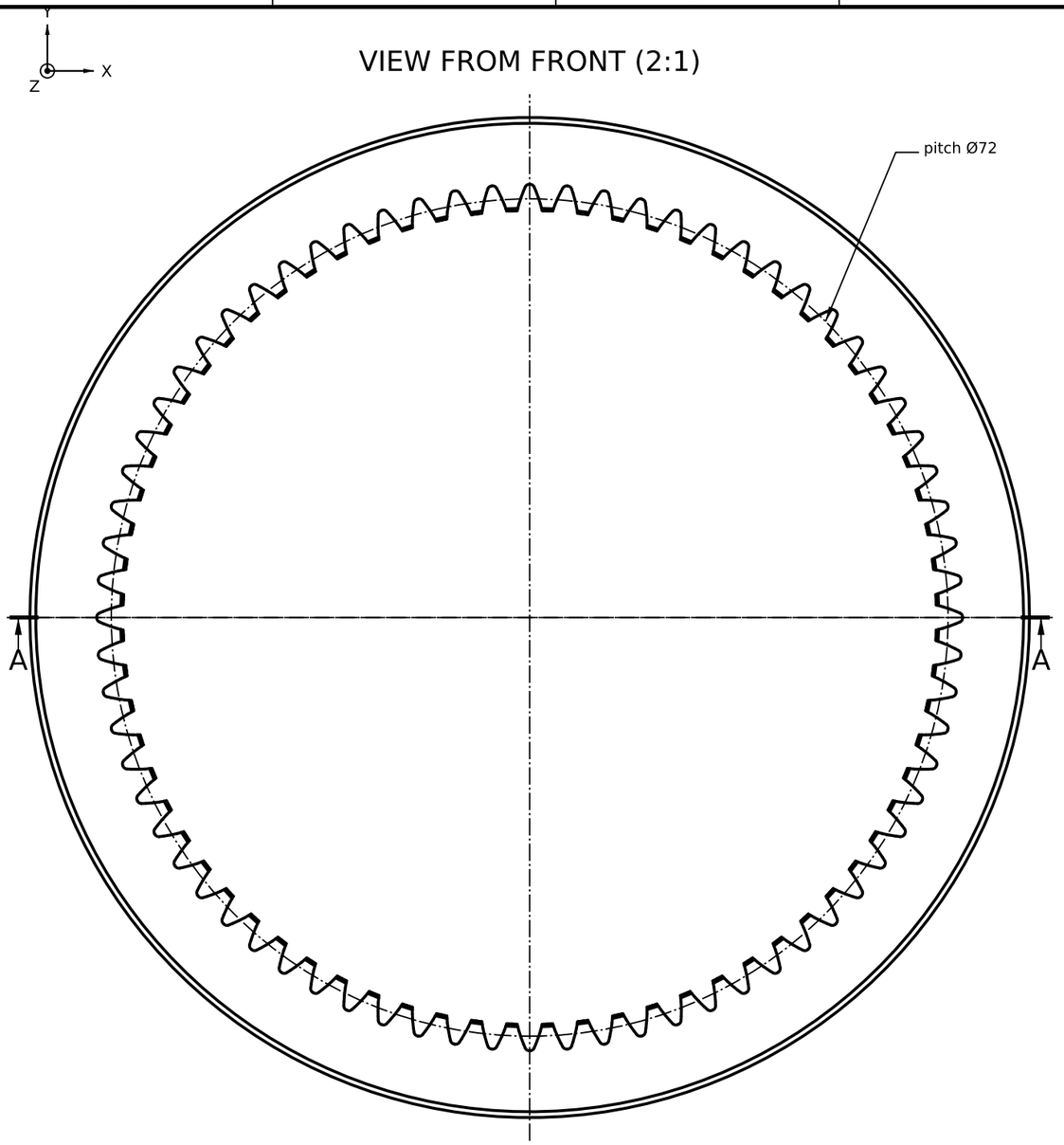
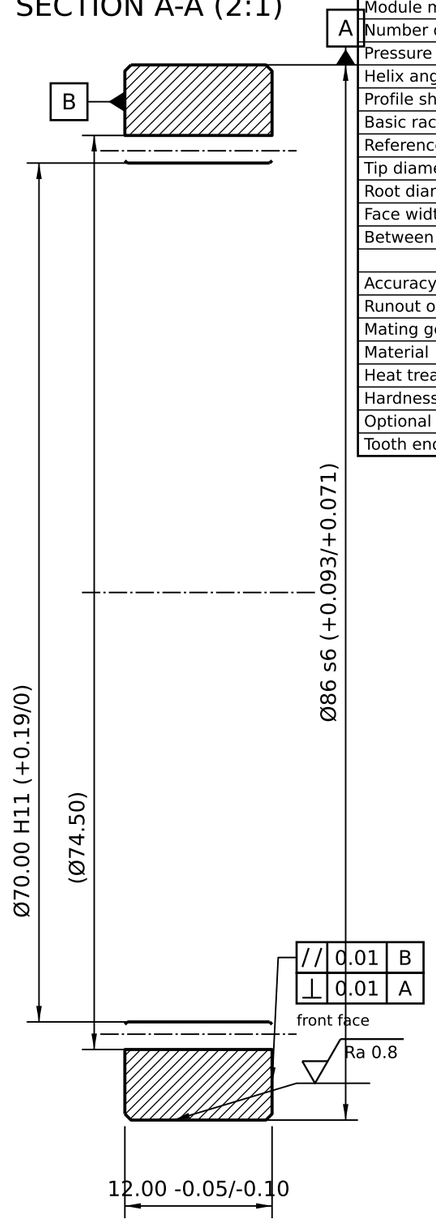
VIEW FROM FRONT (2:1)



SECTION A-A (2:1)



GEAR DATA	
Module m (normal)	1
Number of teeth z	72 (internal)
Pressure angle α	20°
Helix angle β	0° (spur)
Profile shift coefficient x	0
Basic rack	ISO 53 profile A (root radius 0.38)
Reference (pitch) diameter d	72.000
Tip diameter da (inside)	70.00 H11 (+0.19/0)
Root diameter df	74.50 (generated)
Face width b	12.00 -0.05/-0.10
Between 2 pins Ø1.68 Mi	69.703 nom.; make 69.743 / 69.773 (+0.04/+0.07)
	(even z: opposite gaps; space widening for backlash)
Accuracy	ISO 1328-1:2013 grade 7 (shaped)
Runout of teeth (pitch circle) to A	0.015
Mating gears	3 × planet AM-14 (z 27)
Material	34CrNiMo6 +QT (1.6582); AISI 4340 with cert.
Heat treatment	quench and temper before cutting
Hardness	28-34 HRC (about 280-320 HB)
Optional	gas nitriding after cutting: ≥550 HV, Nht 0.2-0.3
Tooth ends	tip chamfer 0.2×45°

NOTES

1. Dimensions in mm. General tolerances ISO 2768-mK. Fits ISO 286-1/-2 (deviations shown in brackets).
2. STEP file governs undimensioned geometry: cnc/step/ring_gear.step (STEVE 02e31a10 rev 49).
3. Edges ISO 13715 -0.3 (break 0.2-0.4 x 45°) unless a chamfer is shown. Remove all burrs, especially in bores and holes.
4. Surface roughness Ra 3.2 µm unless noted (ISO 21920-1).
5. Teeth are hobbed / shaped and ground by the gear shop to the gear data table; the STEP teeth are a reference shape (ISO 53 profile A with the 0.38 m root fillet), not a machining surface.
6. Teeth shaped (Fellows) or, for one or two prototypes, wire-EDM cut. No catalogue m1 72 T internal gear exists.
7. OD Ø86 s6 is pressed into the front flange AM-04 (Ø86 H7) with Loctite 638: 0.036-0.093 interference. The press fit squeezes the teeth inward by about 0.02 mm on the diameter: size Mi for it or check Mi after pressing.
8. OD: 0.5×45° press lead-in chamfer at both ends. Tooth tips: 0.2×45° chamfer at both ends.
9. Inspection: gear report per ISO 1328, Mi, hardness. Supply material certificate.

REV	DESCRIPTION	DATE	BY
B	Space widening for 0.05-0.10 backlash (was 0.03-0.08)	2026-10-17	Claude

PROJECT: **Unknown machine actuator**	DRAWN BY: Claude for macncrash	SOURCE (MODEL GOVERNS): STEVE 02e31a10 rev 49 STEP: cnc/step/ring_gear.step
GENERAL TOLERANCES: ISO 2768-mK (ISO 22081) Edges ISO 13715 -0.3 (0.2-0.4)	PROJECTION: FIRST ANGLE ISO 128 / ISO 5456-2	
FINISH / HEAT TREATMENT: QT 28-34 HRC; gas nitriding optional		UNITS: mm
MATERIAL: 34CrNiMo6 +QT (1.6582)		SCALE: 2:1
TITLE: **Ring gear m1 72 T internal**		QTY PER ACTUATOR: **1**

DRAWING / PART NUMBER	REV	SHEET	SIZE	DATE OF ISSUE
AM-15-ring_gear	**B**	1/1	A3	2026-10-17